

Requirements Table

Requirements Table — Telemedicine Slot Booking & Prescription Portal

Stakeholders / Actors: Patient, Attending Physician

Functional Requirements

FR-001 — Digitally Signed Prescription Generation

Field	Detail
Priority	High
Description	The system shall enable physicians to author and digitally sign structured prescriptions that are automatically compiled into downloadable PDF records.
Acceptance Criteria	Pass: Valid cryptographic signature attached to prescription PDF. Fail: Prescription generated without doctor license validation.
Rationale	Ensures prescriptions are legally valid, tamper-evident, and traceable to a verified, licensed physician — critical for patient safety and regulatory compliance.

FR-002 — Doctor Specialty & Slot Discovery

Field	Detail
Priority	High
Description	The system shall allow patients to search and filter doctors by specialty, availability, and consultation type, and view their open time slots in real time.
Acceptance Criteria	Pass: Search returns only doctors matching the selected specialty with accurate, up-to-date slot availability. Fail: Patient is shown a slot that is already booked or a doctor outside the selected specialty.
Rationale	Patients need an accurate, filterable view of available physicians to make an informed and timely booking decision.

FR-003 — Video Consultation Slot Booking

Field	Detail
Priority	High
Description	The system shall allow patients to book, reschedule, or cancel a video consultation slot with a chosen physician, subject to real-time availability validation.
Acceptance Criteria	Pass: Booking is confirmed only after the system validates the slot is still open at the time of submission. Fail: Two patients are able to book the same slot with the same physician.
Rationale	Prevents double-booking and ensures both patient and physician have a reliable, conflict-free schedule.

FR-004 — Encrypted Tele-Consultation Room Link Generation

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Field	Detail
Priority	High
Description	Upon a confirmed booking, the system shall generate a unique, encrypted video consultation room link and share it with both the patient and physician ahead of the scheduled time.
Acceptance Criteria	Pass: Room link is unique per session, accessible only to the two assigned participants, and expires after the session window. Fail: Room link can be accessed by an unauthenticated or unassigned user.
Rationale	Protects patient privacy and clinical confidentiality during remote consultations, in line with healthcare data protection norms.

FR-005 — Prescription Access & Download

Field	Detail
Priority	Medium
Description	The system shall allow patients to view and download their digitally signed prescription PDFs from their consultation history at any time after issuance.
Acceptance Criteria	Pass: Patient can retrieve and download the correct, signed PDF tied to the correct consultation. Fail: Patient is able to view another patient's prescription record.
Rationale	Gives patients ongoing, self-service access to their medical records for pharmacy use, insurance claims, or personal reference.

Non-Functional Requirements

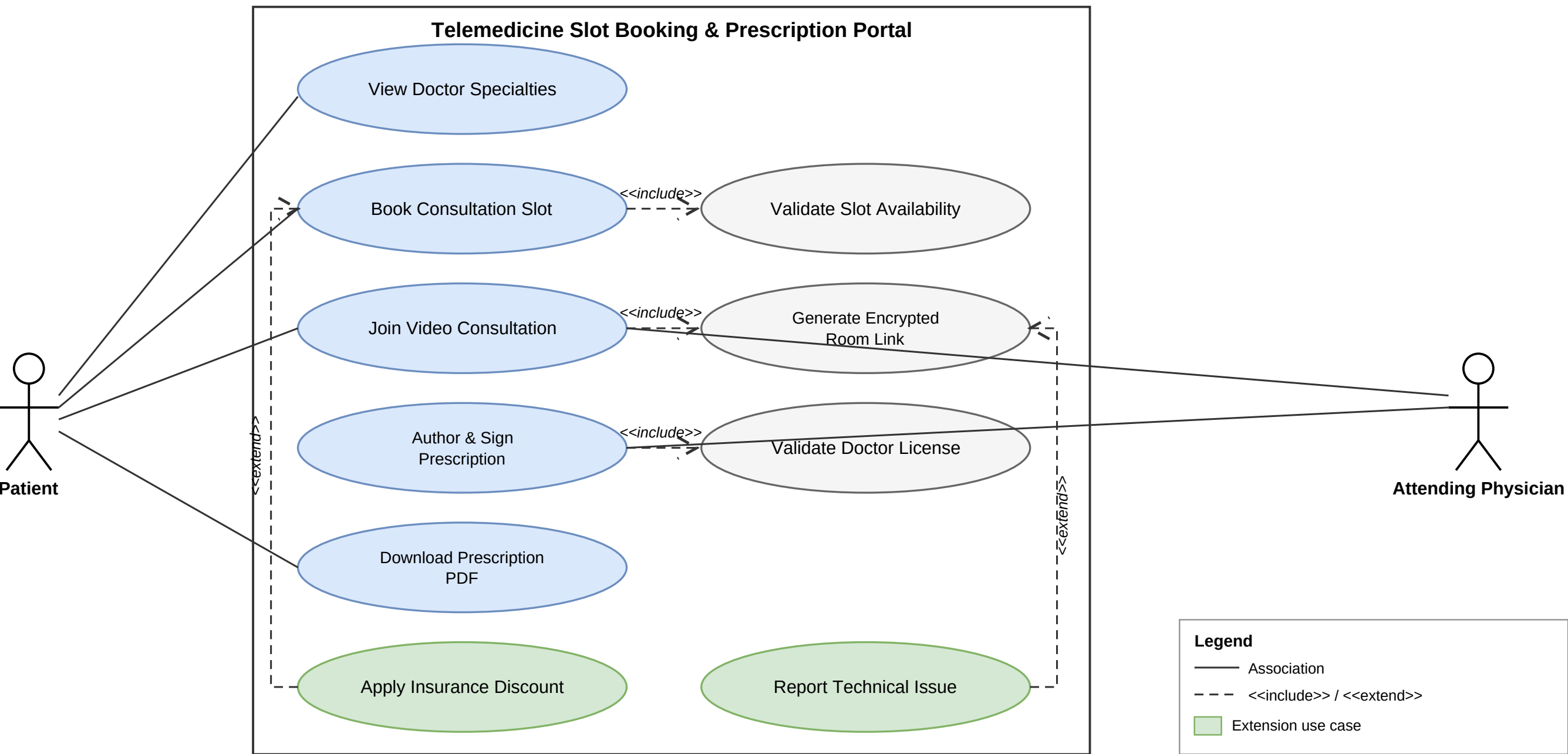
NFR-001 — Data Security in Transit & at Rest

Field	Detail
Type	Performance & Security
Priority	High
Description	All video session metadata and patient clinical notes must be encrypted in transit using TLS 1.3 and at rest with AES-256.
Acceptance Criteria	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Any clinical data transmitted or stored without encryption.
Rationale	Healthcare data is highly sensitive; encryption is a baseline requirement for regulatory compliance (e.g. HIPAA-equivalent standards) and patient trust.

NFR-002 — System Availability & Reliability

Field	Detail
Type	Reliability & Availability
Priority	High
Description	The platform shall maintain at least 99.5% uptime, with automatic failover for video consultation services to prevent dropped or missed appointments.
Acceptance Criteria	Pass: Monthly uptime monitoring confirms $\geq 99.5\%$ availability, with failover activating within 5 seconds of a detected outage. Fail: A scheduled consultation cannot start due to unplanned system downtime.

Rationale	Missed or interrupted consultations directly impact patient care; high availability is essential for a healthcare-critical system.
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Use-Case Flow Specification

Use-Case Flow Specification

Use Case: Book Video Consultation Slot

Actor(s): Patient (primary), System (Slot Availability Service) **Related Use Cases:** «include» Validate Slot Availability · «extend» Apply Insurance Discount

Preconditions

- 1. The patient is authenticated and logged into the portal.
- 2. The patient’s profile has been verified (identity and contact details confirmed).
- 3. At least one physician has published available consultation slots in the system.

Postconditions

- 1. **Success:** A confirmed booking record is created and linked to the patient and the selected physician; the slot is marked unavailable to other patients; a confirmation notification is sent to both the patient and physician.
- 2. **Failure:** No booking record is created; the patient is returned to the slot selection screen with an updated (accurate) list of available slots.

Main Success Scenario

- 1. The patient navigates to the “Book Consultation” section and selects a medical specialty.
- 2. The system displays a list of physicians matching that specialty along with their available slots.
- 3. The patient selects a preferred physician and a specific date/time slot.
- 4. The system triggers the «include» **Validate Slot Availability** use case to confirm the selected slot has not been taken by another patient in the interim.
- 5. The system confirms the slot is available and locks it temporarily for the patient.
- 6. The patient reviews the booking summary (physician name, date, time, consultation fee) and confirms the booking.
- 7. The system processes the booking, permanently reserves the slot, and creates a booking record.
- 8. The system generates and sends a booking confirmation to the patient and physician, including consultation details.
- 9. The use case ends successfully.

Alternate Flow A1: Selected Slot Becomes Unavailable

Trigger: At step 4/5, the slot validation check finds the selected slot has already been booked by another patient (race condition).

- 1. The system detects the slot is no longer available during validation.
- 2. The system displays a message to the patient: *“This slot was just booked by another patient. Please select a different time.”*
- 3. The system refreshes and displays the physician’s current list of available slots.
- 4. The patient selects a new slot from the updated list.
- 5. The flow resumes at **Main Success Scenario, Step 4** (validate the newly selected slot).

*(If the patient chooses not to select a new slot, the use case ends without a booking being created —
Postcondition: Failure.)*